

Academic-v2 Manuscript Revision Guide

Code-to-paper audit, updated 11 September 2026

How to use this file

Red blocks preserve claims or methods currently present in `cas-sc-template.tex`. Cyan blocks are proposed replacements grounded in the executed academic-v2 scripts and the validated result files. This document does not silently delete earlier review markup. Exact normalized macro F_1 is the confirmatory metric; containment is a pre-declared secondary boundary-relaxed metric; lexical and semantic matching remain diagnostics.

1 Changes required before the academic-v2 results can enter the paper

Area	Why the current paper is stale	Required direction
Primary metric	The paper calls containment primary, while every academic-v2 HPO run maximized <code>eval_exact_f1</code> .	Make exact normalized macro F_1 primary and containment secondary.
Canonicalization	The paper says longer phrases are replaced by nested shorter phrases. That rule was removed.	State exact deduplication only; retain semantically distinct nested phrases.
Matching	The paper describes prediction-order greedy allocation and character-substring containment.	State deterministic maximum-cardinality matching at each cascade level and contiguous-token containment.
Splitting	The figure says “Grouped split”, but the generator assigns IDs independently and intentionally does not group equal text under different IDs.	Use “identifier-level stratified 40/10/50 split”. Disclose five unavoidable IT-all prompt-development IDs in test and the sensitivity analysis.
BERT	The paper says description-only input, unweighted loss, and inconsistent train/test input.	State headline plus newline plus description for all partitions and weighted cross entropy from train BIO counts.
BART	The paper says 96 target tokens and every chunk receives the full target list.	State 512 target/generation tokens and source-visible chunk targets with deterministic fallback.
HPO reproducibility	The paper says Optuna defaults.	State <code>TPESampler(seed=13)</code> , RNG reset before every model initialization, 20 trials, and one final seed.
Results	Tables and discussion use the legacy scorer/split.	Replace them with the tables below and change the winner claims.

Controls	The legacy before–after values did not use the final prompt-safe test sets and scorer.	Replace them with the 36 validated task-unadapted controls and centralized before–after table.
LLM provenance	Parsed workbooks survive, but exact provider snapshots, raw responses, and retry histories do not.	Report model families conservatively and list the missing provenance as a limitation.
Inter-annotator agreement	The table inherits a comma-splitting parser, nested-phrase canonicalization, and coverage logic that differ from the method described in the paper.	Recalculate the table with bracket-aware parsing, exact-only deduplication, and the academic-v2 exact-first one-to-one matcher.

2 Metric framing and HPO objective

CURRENT TEXT, KEPT FOR REVIEW

Containment is treated as the primary measure. The boundary-relaxed containment macro F_1 served as the validation objective and the primary cross-configuration metric. Optuna returned exact matches plus containment matches.

PROPOSED ACADEMIC-V2 TEXT

Exact normalized macro F_1 was the validation objective and the primary confirmatory metric. Boundary-relaxed containment macro F_1 was reported as a secondary measure of boundary sensitivity. Lexical and semantic matching were supplementary diagnostics and did not affect hyperparameter selection. For each of the 36 supervised configurations, Optuna executed 20 trials with a TPE sampler seeded at 13. The random-number generators were reset immediately before every trial-specific model initialization and before final training.

3 Normalization and one-to-one matching

CURRENT TEXT, KEPT FOR REVIEW

Exact duplicates and longer phrases containing an already retained shorter skill were canonicalized to a single item. Matching used a greedy cascade in prediction order, and containment accepted either normalized string when it contained the other.

PROPOSED ACADEMIC-V2 TEXT

Normalization applied Unicode NFKC normalization, whitespace and dash harmonization, delimiter cleanup, case-insensitive comparison keys, and normalization-aware exact deduplication. It did not replace a longer skill with a nested shorter skill. Predictions and gold items were matched one-to-one in a fixed cascade. At each level, a deterministic maximum-cardinality bipartite matching allocated all eligible pairs before the next level began. The levels were exact equality, contiguous-token containment, lexical similarity, and semantic similarity. Hence, prediction order could not consume a gold item that had an exact match elsewhere.

4 Inter-annotator agreement

CURRENT TEXT, KEPT FOR REVIEW

The paper states that both annotator lists used the same surface processing as model output, that one-to-one exact and containment pairs produced the current table, that containment was the primary reliability measure, and that the exact–containment difference reflected boundary variation rather than skill identification disagreement. It further interprets values between 0.7 and 1 as “substantial consistency.”

PROPOSED ACADEMIC-V2 TEXT

To retain one shared metric definition, Section Task Formulation and Key Evaluation Approaches should define two generic normalized mention sets $\mathcal{A}$ and $\mathcal{B}$ and let T be the number of one-to-one pairs assigned under a named matching rule. Equation 1 then applies without treating either human annotator as ground truth. For model evaluation, $\mathcal{A}$ is the prediction set and $\mathcal{B}$ is the gold set; for inter-annotator agreement, the labels “annotator 1” and “annotator 2” are only a computational orientation, and the resulting F_1 is symmetric.

Pairwise mention-set agreement was recalculated on the 32 shared software-developer advertisements and 35 shared journalist advertisements. Annotator cells were parsed with the bracket-aware delimiter rule used during dataset processing. Both lists received the same Unicode, whitespace, punctuation, technology-spelling, and exact-deduplication normalization used by the academic-v2 scorer. Prediction-only fragment stopwords, blacklists, prefixes, and allowlists were not applied to human annotations. Exact pairs were assigned first by deterministic maximum-cardinality bipartite matching; contiguous-token containment pairs were then assigned only among the remaining items. Counts were pooled over each domain–label overlap sample before precision, recall, and micro F_1 were calculated.

Exact agreement is the confirmatory column and containment is a secondary view of boundary sensitivity. Their difference is consistent with boundary variation but can also include differences in semantic scope, so it should not be attributed exclusively to boundaries. Pairwise mention-set F_1 measures observed overlap and is not a chance-corrected coefficient such as Cohen’s kappa; therefore, kappa-style qualitative bands such as “substantial” should not be assigned to these values.

Job category	Skill label	Records	Exact	Boundary-relaxed containment
Software developers	Hard skills	32	0.692	0.882
Software developers	Soft skills	32	0.676	0.889
Software developers	Distinct skills	32	0.667	1.000
Software developers	Must-have skills	32	0.652	0.914
Software developers	Nice-to-have skills	32	0.701	0.948
Journalists	Hard skills	35	0.489	0.792
Journalists	Soft skills	35	0.763	0.844
Journalists	Distinct skills	35	0.407	0.712
Journalists	Must-have skills	35	0.620	0.856
Journalists	Nice-to-have skills	35	0.791	0.977

5 Data processing and splitting

CURRENT TEXT, KEPT FOR REVIEW

The manuscript describes a grouped 40/10/50 split and states that grouping prevents the same advertisement from crossing partitions. It also reports the pre-parser-correction label counts.

PROPOSED ACADEMIC-V2 TEXT

The corrected corpora contain 315 software-developer advertisements, 322 journalist advertisements, and 637 records in the merged corpus, each with a unique identifier. All five annotation fields were parsed with a bracket-aware delimiter rule so commas inside matched brackets were retained inside one skill phrase. For each domain-skill configuration, identifiers were assigned to 40% train, 10% validation, and 50% test using seed 13. The balancing objective used input length, target-item count, target-word count, and quantile indicators; all-skills splits additionally balanced the five component-label counts, and merged splits balanced occupation membership. IDs used during prompt development were kept out of test whenever feasible. This was feasible for 17 configurations; IT-all necessarily retained five such IDs because the clean non-prompt pool was smaller than the fixed test size. Excluding those five IDs changed macro F_1 by at most 0.0033 and did not change the principal domain conclusions. The implementation groups by identifier, not by normalized text, so the paper must not claim text-level grouping.

Domain	Ads	IDs	Hard	Soft	Distinct	Must	Nice	All
Software developers	315	315	4221	2343	179	2291	907	6643
Journalists	322	322	4853	1825	534	6644	434	7171
Merged	637	637	9074	4168	713	8935	1341	13814

The component columns count annotated list items before cross-label merging; “All” counts the exact-normalized union stored in `all_skills_list`. Therefore the component counts are not intended to sum to the All column.

6 BERT implementation

CURRENT TEXT, KEPT FOR REVIEW

Training and validation use description only, final test combines headline and description, and ordinary cross-entropy loss is minimized.

PROPOSED ACADEMIC-V2 TEXT

The same model input was used in training, validation, and test: the stripped headline and description joined by a newline. Annotated strings were weakly aligned to case-insensitive contiguous source spans, tokenized into windows of at most 512 subwords with stride 128, and encoded as B-SKILL, I-SKILL, and O. All 18 BERT configurations used weighted token cross entropy. Weights were computed only from the training partition as $w_c = \min(10, \max(1, \sqrt{N_O/N_c}))$, with $w_O = 1$ and padding ignored. Validation and test decoding used evaluation mode; no CRF was used.

7 BART implementation

CURRENT TEXT, KEPT FOR REVIEW

Inputs and targets are truncated to 1,024 and 96 tokens. Each long-text chunk is paired with the complete advertisement-level target list. Training uses the description, while validation and test combine headline and description.

PROPOSED ACADEMIC-V2 TEXT

The same headline–description input was used for all partitions. Inputs longer than 1,000 tokenizer tokens were divided into overlapping chunks with a 160-token overlap. Each gold skill was assigned to every chunk containing its normalized contiguous token sequence; if no chunk contained it, the skill was assigned deterministically to the chunk with the largest token-set overlap. The serialized target and generated continuation were limited to 512 tokens. Validation used beam width two; final test generation used beam width six and a no-repeat trigram constraint. Chunk predictions were merged and exactly deduplicated before the shared prediction filter and scorer were applied.

8 Results that replace the stale tables

Primary: exact normalized macro F_1

Model	Software developers	Journalists	Merged
fine-tuned Swedish BERT	0.438	0.430	0.454
fine-tuned Swedish BART	0.333	0.315	0.355
prompted GPT	0.429	0.317	0.361
prompted Gemini	0.442	0.349	0.382
prompted Claude	0.410	0.293	0.342

Secondary: boundary-relaxed containment macro F_1

Model	Software developers	Journalists	Merged
fine-tuned Swedish BERT	0.621	0.687	0.670
fine-tuned Swedish BART	0.489	0.476	0.528
prompted GPT	0.620	0.563	0.587
prompted Gemini	0.652	0.590	0.615
prompted Claude	0.637	0.557	0.592

Under both metrics, Gemini leads software developers and fine-tuned BERT leads journalists and the merged evaluation. Under exact matching the respective domain scores are 0.442, 0.430, and 0.454; under containment they are 0.652, 0.687, and 0.670. The IT exact margin is only 0.004, so it must be described as a descriptive result from one seed rather than statistically established superiority.

Exact macro F_1 by skill label

Skill label	BERT	BART	GPT	Gemini	Claude
All skills	0.523	0.377	0.433	0.445	0.411
Hard skills	0.473	0.326	0.340	0.348	0.335
Soft skills	0.523	0.478	0.412	0.469	0.431
Distinct skills	0.256	0.180	0.212	0.202	0.177
Must-have skills	0.460	0.338	0.290	0.353	0.297
Nice-to-have skills	0.408	0.305	0.524	0.528	0.437

Containment macro F_1 by skill label

Skill label	BERT	BART	GPT	Gemini	Claude
All skills	0.776	0.569	0.670	0.670	0.656
Hard skills	0.706	0.514	0.566	0.582	0.577
Soft skills	0.726	0.637	0.616	0.679	0.651
Distinct skills	0.466	0.320	0.430	0.422	0.372
Must-have skills	0.667	0.499	0.453	0.542	0.518
Nice-to-have skills	0.616	0.446	0.804	0.819	0.799

9 Winner claims

CURRENT TEXT, KEPT FOR REVIEW

Fine-tuned BERT led 11 of 18 cells; prompted models led seven; GPT led distinct-skill extraction for journalists and merged data; prompted models were strongest on both distinct and nice-to-have skills on average.

PROPOSED ACADEMIC-V2 TEXT

Under the primary exact metric, BERT led 12 of 18 domain-skill cells, Gemini led three, GPT led two, and BART led one; Claude led none. Under the secondary containment metric, BERT led 13 cells, Gemini four, and Claude one; GPT and BART led none. Thus, BERT led all-skills and hard-skills in every domain under both metrics, but the exact and containment views differ for some label-specific cells. Averaged across domains, BERT had the highest exact and containment score for distinct skills, whereas Gemini had the highest score for nice-to-have skills. The supported higher-level conclusion is narrower: prompted-model wins under containment were concentrated on distinct or nice-to-have cells (four of five prompted wins), not that prompted models were uniformly best on both labels.

10 Task-unadapted controls

CURRENT TEXT, KEPT FOR REVIEW

The abstract, Results, Discussion, and Conclusion quantify large before-after fine-tuning gains using the legacy controls and state that the current academic-v2 models improve by those amounts.

PROPOSED ACADEMIC-V2 TEXT

The academic-v2 release contains 36 fine-tuned runs, 36 task-unadapted controls, and 54 aligned prompted-model configurations. Every control matches the corresponding fine-tuned run in canonical test-identifier order, gold labels, normalization, prediction filtering, and centralized scoring.

The BERT control applies a fixed-seed random BIO classification head to the pretrained encoder; the BART control uses the pretrained denoising checkpoint without task-specific updates. Neither control reads train or validation data. They are therefore task-unadapted controls, not architecture-matched zero-shot systems. Averaged across the 18 domain–skill configurations, exact normalized macro F_1 increased from 0.034 to 0.440 for BERT and from 0.00005 to 0.334 for BART. Fine-tuning improved this primary metric in all 36 paired configurations.

Family	Domain	Exact macro F_1			Containment macro F_1		
		Control	Fine-tuned	Δ	Control	Fine-tuned	Δ
BERT	Software developers	0.041	0.438	+0.398	0.128	0.621	+0.494
BERT	Journalists	0.028	0.430	+0.402	0.159	0.687	+0.529
BERT	Merged	0.034	0.454	+0.420	0.146	0.670	+0.524
BART	Software developers	0.000	0.333	+0.333	0.052	0.489	+0.438
BART	Journalists	0.000	0.315	+0.315	0.062	0.476	+0.413
BART	Merged	0.000	0.355	+0.354	0.055	0.528	+0.473

11 Filtering, unsupported output, and fairness

The shared filter removed 4.71% of raw BERT items, 0.01% of BART items, and approximately 0.05% of prompted-model items. This difference must be disclosed even though every system used the same rule files. Unsupported-item rates were structurally zero for source-bound BERT, 4.02% for BART, 0.48% for GPT, 1.00% for Gemini, and 5.69% for Claude. These rates are useful diagnostics but are not a like-for-like architectural score because BERT cannot generate text outside the source.

All five systems share test identifiers, gold lists, normalization, filtering, and scoring. This establishes arithmetic comparability, not equivalence of training resources: BERT and BART receive supervised parameter updates, while the commercial systems use prompts developed with labeled examples. The paper must call this a comparison of adaptation and resource regimes, not a fair architecture-only experiment.

12 Limitations that should remain explicit

Only one training/data seed was run, and retained commercial outputs contain no repeated API calls. Small score differences therefore remain descriptive. The surviving LLM workbooks do not preserve exact provider snapshot IDs, raw API responses, retry histories, or original format failures. Twenty-two unique IDs show whitespace or punctuation tokenization differences between retained LLM text and canonical text, but all differences disappear after removing whitespace and punctuation. Per-skill datasets contain only advertisements with at least one annotation for that label; performance on genuinely negative advertisements is therefore not estimated. Finally, IDs are split independently, so equal text under different IDs is not prevented from crossing partitions.

13 Files that substantiate this guide

The primary audit tables are `all_models_all_metrics.csv`, `all_models_per_ad_metrics.csv`, `neural_recalculation_audit.csv`, `alignment_audit.csv`, `claim_audit.csv`, `fairness_audit.csv`, and `manuscript_method_consistency.csv`. Control validation and centralized scores are recorded in `baseline_output_validation.csv` and `task_unadapted_all_metrics.csv`; the paired analysis is in `before_after_by_configuration.csv`. Inter-annotator calculations and their manuscript-value comparison are in `reliability_summary.csv` and `manuscript_table_comparison.csv`. The release manifest records the SHA-256 checksum and provenance role of every packaged file.